H2 Math Summary

Example Questions

Maximus Pung

Contents

1 Systems of Linear Equations **2**

Question 1.1 [MI 2025 Prelim P1 Q4] — Forming equations from a table 2

Question 1.2 [SAJC 2025 Prelim P1 Q1] — Unknown coefficients of a polynomial 3

Question 1.3 [HCI 2025 Prelim P1 Q2] — Unknown constants in a recurrence relation 4

Question 1.4 [SAJC 2025 Prelim P1 Q3] — More equations than variables 5

Question 1.5 [TJC 2022 Prelim P2 Q1] — Fewer equations than variables with an inequality 6

2 Inequalities **7**

Question 2.1 [RVHS 2026 Prelim P1 Q2] — Graphical method and sign test 7

Question 2.2 [ACJC 2026 Prelim P1 Q1] — Sign test and $\frac{1}{x}$ substitution 8

Question 2.3 [SAJC 2025 Prelim P2 Q1] — Modulus inequality and substitution 9

Question 2.4 [SAJC 2019 Prelim P1 Q2] — Modulus graphs and area 10

3 Graphing Techniques **11**

Question 3.1 [VJC 2025 P1 Q5] — Periodic piecewise graph and hyperbola asymptotes 11

Question 3.2 [VJC 2025 P1 Q8] — Reciprocal graph turning point and derivative graph 13

Question 3.3 [SAJC 2025 P1 Q7] — Rational graph and line intersection with area 15

Question 3.4 [DHS 2025 P1 Q5] — Graph of $y = \mathrm{f}(x)$ from $y = \mathrm{f}'(x)$ 16

Question 3.5 [DHS 2025 P1 Q6] — Parametric curve with odd symmetry 18

Question 3.6 [DHS 2025 P1 Q8] — Ellipse from a parametric complex locus 20

Question 3.7 [DHS 2025 P2 Q1] — Rational graph with oblique asymptote and concavity 21

1 Systems of Linear Equations

1.1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	Discounts given for each package			Total price after the discount
	Blend A	Blend B	Blend C	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		Comments
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

1.2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		**Comments**
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

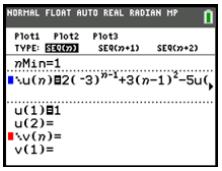
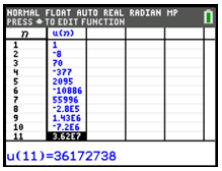
1.3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		**Comments**
[2]	$n = 1$ $x_2 = a(-3)^1 + b(1)^2 + cx_1$ $-8 = -3a + b + c \quad ---(1)$ $n = 2$ $x_3 = a(-3)^2 + b(2)^2 + cx_2$ $70 = 9a + 4b - 8c \quad ---(2)$ $n = 3$ $x_4 = a(-3)^3 + b(3)^2 + cx_3$ $-377 = -27a + 9b + 70c \quad ---(3)$ Using GC, $a = 2,\ b = 3,\ c = -5$ $x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$ Using GC, $x_{10} = -7\,210\,868$ $x_{11} = 36\,172\,738$   	

1.4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

Solution		**Comments**
(i) **[2]**	$1-2\lambda = 2\mu$ $---(1)$ $2+\lambda = 5-2\mu$ $---(2)$ $2\lambda = -8+3\mu$ $---(3)$ Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$ Sub into (3): $\text{LHS} = 2(-2) = -4$ $\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$ $\text{LHS} \neq \text{RHS},$ no solution $\therefore l_1$ and l_2 do not intersect $\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$ $\therefore l_1$ and l_2 are not parallel $\therefore l_1$ and l_2 are skew (shown)	

1.5 A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

Age group	Discounted ticket price
$\leqslant 12$ years	\$6
13 to 49 years	\$9
$\geqslant 50$ years (senior citizen)	\$4

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

[TJC 2022 Prelim P2 Q1]

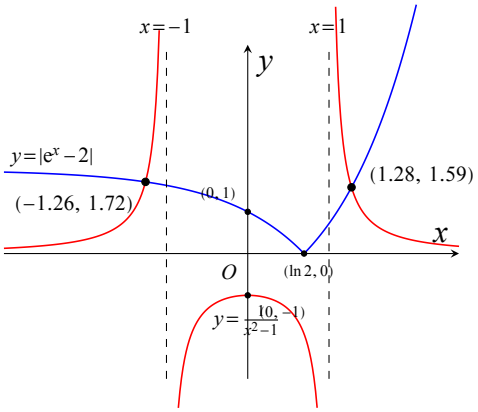
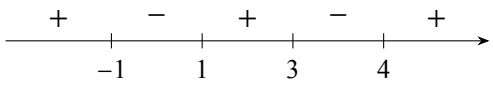
Solution		**Comments**
[4]	Let x, y, z be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively. $\left.\begin{aligned} x+y+z=100 \quad &---(1) \\ 6x+9y+4z=700 \quad &---(2) \end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$ $z \geqslant 35$ $\because x \geqslant 0$ $\frac{200}{3}-\frac{5}{3}z \geqslant 0$ $z \leqslant 40$ $\therefore 35 \leqslant z \leqslant 40$ x, y, z are non-negative integers. Using GC table: $z=37 \Rightarrow x=5,\ y=58$ $z=40 \Rightarrow x=0,\ y=60$ $\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$	

2 Inequalities

2.1 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

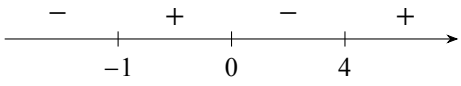
Solution		Comments
(a) **[3]**	 Using GC, intersections at $x=-1.2581$, $x=1.2770$ $\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$	
(b) **[3]**	$\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$ $\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$ $\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$ $\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$  $\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$	

2.2 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

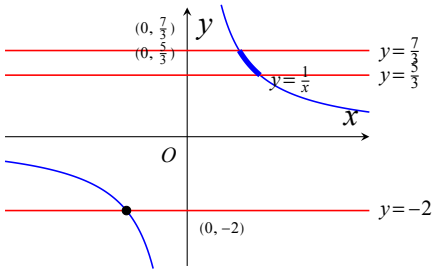
[ACJC 2026 Prelim P1 Q1]

Solution		**Comments**
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\frac{x(5x+9)}{x^2+2x+5} > x$ $5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1 \text{ or } 0<x<4 \quad ---(2)$ $(1) \cap (2)$: $\therefore x<-1 \text{ or } 0<x \leqslant 3$	

2.3 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

[SAJC 2025 Prelim P2 Q1]

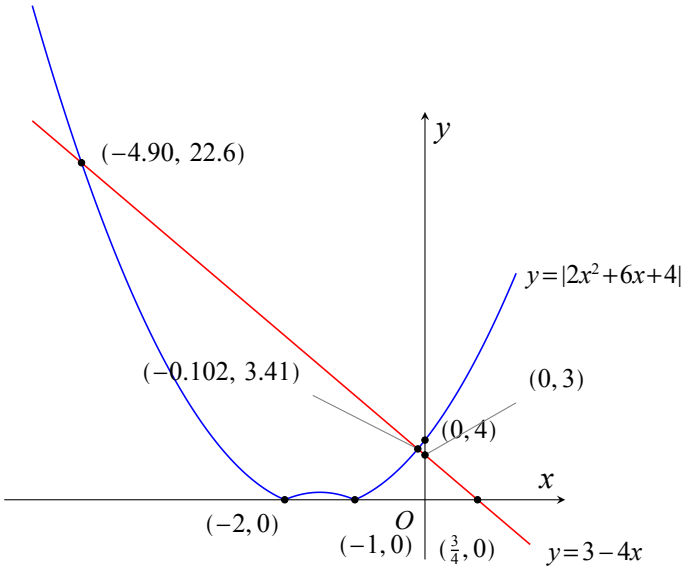
Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2$: $0 \leqslant 0$ (true) $x \neq -2$: $\|x+2\| > 0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$ $3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	

2.4 **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

(ii) Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

[SAJC 2019 Prelim P1 Q2]

Solution	Comments
(i) **[4]**	



$$3-4x=2x^2+6x+4$$

$$2x^2+10x+1=0$$

$$x=\frac{-5\pm\sqrt{23}}{2}$$

$$\therefore x<\frac{-5-\sqrt{23}}{2} \text{ or } x>\frac{-5+\sqrt{23}}{2}$$

(ii) **[3]**

$$\text{Area}=\int_{-3}^{-1}(3-4x-|2x^2+6x+4|)\,dx$$

$$=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$$

$$=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$$

$$=\left[-\tfrac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\tfrac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$$

$$=\left(\tfrac{16}{3}-20+2\right)-(18-45+3)+\left(-\tfrac{2}{3}+1-7\right)-\left(-\tfrac{16}{3}+4-14\right)$$

$$=20$$

3 Graphing Techniques

3.1 The function f is given by

$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of x.

(a) On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

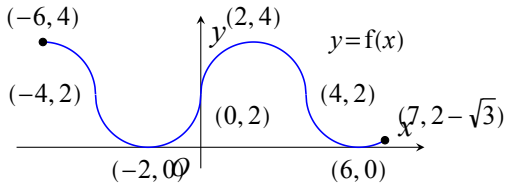
(b) Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the x-axis and the y-axis. [1]

The curve C has equation $\dfrac{(y-3)^2}{a^2}-(x+2)^2=1$, where a is a positive real constant.

(c) State the equations of the asymptotes of C in terms of a. [1]

(d) Determine the range of values of a if there is at most one intersection between C and the graph of $y=\mathrm{f}(x)$. [2]

[VJC 2025 P1 Q5]

Solution		Comments
(a) **[3]**		
(b) **[1]**	$\text{Area} = 2\times 4 + \frac{1}{2}\pi(2^2)$ $= 8 + 2\pi$	
(c) **[1]**	$\frac{(y-3)^2}{a^2} = (x+2)^2$ $y - 3 = \pm a(x+2)$ $y = 3 + a(x+2)$ and $y = 3 - a(x+2)$	
(d) **[2]**	C is a hyperbola with centre $(-2, 3)$ and vertices $(-2, 3+a)$, $(-2, 3-a)$ When $a = 3$, exactly one intersection at $(-2, 0)$ $\therefore a \geqslant 3$	

3.2 **(a)** A curve C with equation $y=\mathrm{f}(x)$ undergoes in succession, the following transformations.

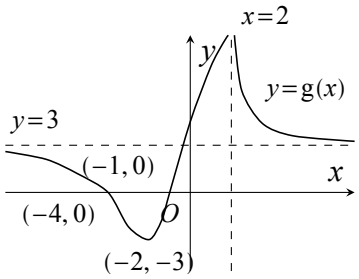
A: A reflection in the x-axis.

B: A stretch parallel to the x-axis with scale factor $\frac{1}{2}$, with the y-axis invariant.

The resulting curve has equation $y=ax^2+\dfrac{b}{x}$, where a and b are real constants.

Given that $\left(-1, \frac{1}{3}\right)$ is a turning point of $y=\dfrac{1}{\mathrm{f}(x)}$, find the values of a and b and state the equation of C. [5]

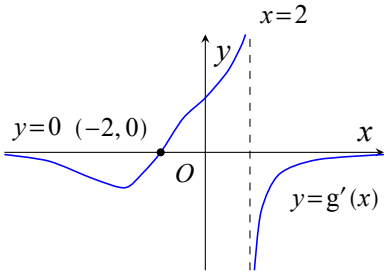
(b) The diagram below shows the curve of $y=\mathrm{g}(x)$. The curve has a minimum point at $(-2,-3)$ and crosses the x-axis at $(-1,0)$ and $(-4,0)$. The line $x=2$ is the vertical asymptote and the line $y=3$ is the horizontal asymptote.



(i) Sketch the graph of $y=\mathrm{g}'(x)$, labelling the coordinates of all relevant point(s) and state the equations of any asymptotes. [2]

(ii) Find the area of the region bounded by the graph of $y=\mathrm{g}'(x)$, the lines $x=-4$, $x=-2$ and the x-axis. [2]

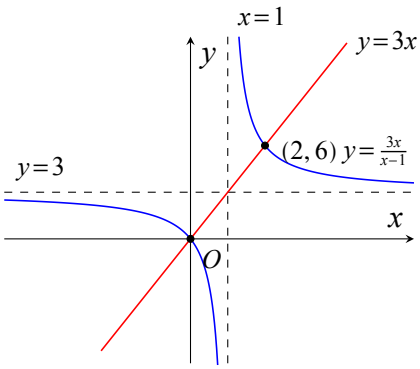
[VJC 2025 P1 Q8]

Solution		Comments
(a) **[5]**	$y=ax^2+\dfrac{b}{x}$	
	$\xrightarrow{B^{-1}:\ x\to\frac{x}{2}}\quad y=a\left(\dfrac{x}{2}\right)^2+\dfrac{b}{\frac{x}{2}}=\dfrac{ax^2}{4}+\dfrac{2b}{x}$	
	$\xrightarrow{A^{-1}:\ y\to -y}\quad y=-\dfrac{ax^2}{4}-\dfrac{2b}{x}=\mathrm{f}(x)$	
	$\left(-1,\frac{1}{3}\right)$ on $y=\frac{1}{\mathrm{f}(x)}\ \Rightarrow\ (-1,3)$ is a turning point on $y=\mathrm{f}(x)$	
	$-\dfrac{a}{4}+2b=3$	$---(1)$
	$\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{1}{2}ax+\dfrac{2b}{x^2}$	
	$x=-1,\ \dfrac{\mathrm{d}y}{\mathrm{d}x}=0:\quad \dfrac{1}{2}a+2b=0$	$---(2)$
	Using GC, $a=-4,\ b=1$	
	$\therefore C:\ y=x^2-\dfrac{2}{x}$	
(b)(i) **[2]**		
	Asymptotes: $x=2,\ y=0$	
(b)(ii) **[2]**	Required area $=-\displaystyle\int_{-4}^{-2}\mathrm{g}'(x)\,\mathrm{d}x$	
	$=-\left[\mathrm{g}(x)\right]_{-4}^{-2}$	
	$=-\left[\mathrm{g}(-2)-\mathrm{g}(-4)\right]$	
	$=-\left[-3-0\right]$	
	$=3$ units^2	

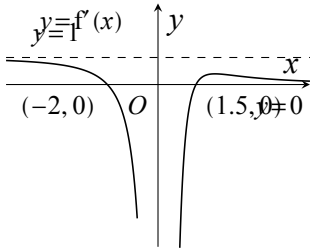
3.3 (a) (i) On a single diagram, sketch the graphs of $y=\dfrac{3x}{x-1}$ and $y=3x$, stating the equation(s) of any asymptote(s) and the coordinates of points of intersection. [4]

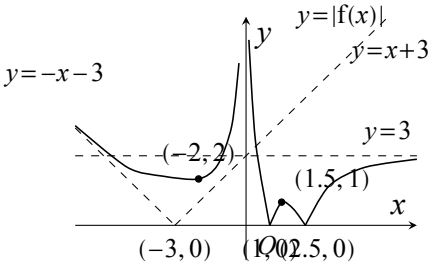
(ii) Hence find the area bounded by the curve $y=\dfrac{3x}{x-1}$ and the lines $y=3x$ and $y=8$. [2]

[SAJC 2025 P1 Q7]

Solution		Comments
(a)(i) **[4]**	 $\dfrac{3x}{x-1}=3x$ $3x=3x(x-1)$ $3x(x-2)=0$ $x=0$ or $x=2$ $\therefore$ intersections: $(0,0)$, $(2,6)$ Asymptotes: $x=1$, $y=3$	
(a)(ii) **[2]**	$y=\dfrac{3x}{x-1} \Rightarrow x=\dfrac{y}{y-3}$ $y=3x \Rightarrow x=\dfrac{y}{3}$ $\text{Area}=\int_6^8 \left(\dfrac{y}{3}-\dfrac{y}{y-3}\right)\,\text{d}y$ $=1.13 \text{ units}^2$ (3 s.f.)	

3.4 The graphs of $y=\mathrm{f}'(x)$ and $y=|\mathrm{f}(x)|$ are shown below.





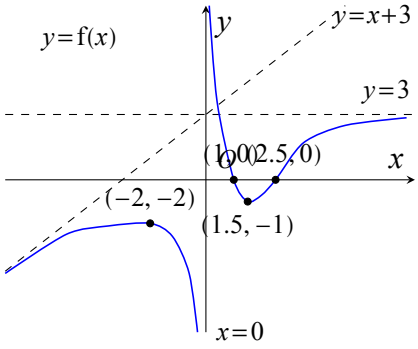
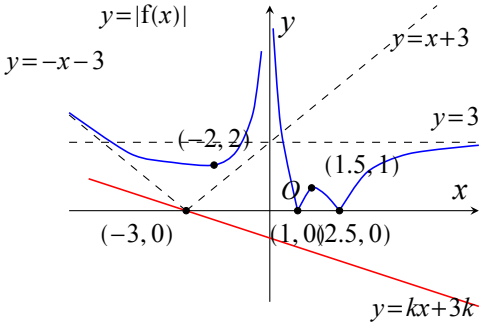
(a) State the nature of all turning point(s) of the graph of $y=\mathrm{f}(x)$. [2]

(b) State the range of values of x where f is decreasing. [2]

(c) Sketch the graph of $y=\mathrm{f}(x)$ indicating clearly the equations of the asymptotes, coordinates of the turning point(s) and the intersections with the axes. [3]

(d) On the copy of the graph of $y=|\mathrm{f}(x)|$ in the Printed Answer Booklet, sketch and label a line $y=kx+3k$, where k is a constant. Hence state the range of values of k for which there is no real solution to the equation $|\mathrm{f}(x)|=kx+3k$. [2]

[DHS 2025 P1 Q5]

Solution		Comments
(a) **[2]**	$x=-2$: $f'(x)$ changes from + to − $\Rightarrow$ maximum point $x=1.5$: $f'(x)$ changes from − to + $\Rightarrow$ minimum point	
(b) **[2]**	$f'(x) \leqslant 0$ $-2 \leqslant x < 0$ or $0 < x \leqslant 1.5$	
(c) **[3]**	 Asymptotes: $x=0$, $y=3$, $y=x+3$	
(d) **[2]**	 $y=kx+3k=k(x+3)$ passes through $(-3, 0)$ $\therefore\ -1 \leqslant k < 0$	

3.5 A curve C is defined parametrically by

$$x = a\tan t, \qquad y = a\sec^2 t\sin t, \qquad -\frac{\pi}{4} < t < \frac{\pi}{4},$$

where a is a positive constant.

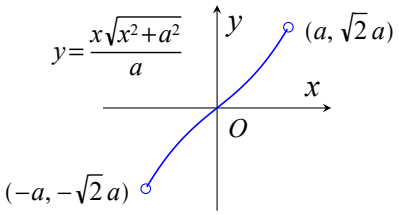
(a) Show that $\dfrac{\tan t}{\sqrt{1+\tan^2 t}} = \sin t$. [1]

(b) By using part **(a)** or otherwise, find the cartesian equation of C in the form $y = \mathrm{f}(x)$, simplifying your answer. [2]

(c) Show that $\mathrm{f}(-x) = -\mathrm{f}(x)$. Hence sketch C. [2]

(d) Find the exact area of the region bounded by C, the x-axis and the lines $x = -A$ and $x = A$, where $0 < A < a$, leaving your answer in terms of A and a. [3]

[DHS 2025 P1 Q6]

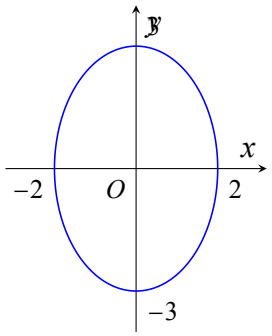
Solution		Comments
(a) **[1]**	$\text{LHS} = \dfrac{\tan t}{\sqrt{1+\tan^2 t}} = \left(\dfrac{\sin t}{\cos t}\right)\left(\dfrac{1}{\sec t}\right)$ $= \left(\dfrac{\sin t}{\cos t}\right)(\cos t)$ $= \sin t = \text{RHS (shown)}$	
(b) **[2]**	$x = a\tan t \Rightarrow \tan t = \dfrac{x}{a}$ $y = a\sec^2 t \sin t$ $= a\left(\tan^2 t+1\right)\left(\dfrac{\tan t}{\sqrt{1+\tan^2 t}}\right)$ $= a\left(\left(\dfrac{x}{a}\right)^2+1\right)\left(\dfrac{\frac{x}{a}}{\sqrt{1+\left(\frac{x}{a}\right)^2}}\right)$ $= x\sqrt{1+\left(\dfrac{x}{a}\right)^2}$ $= \dfrac{x\sqrt{x^2+a^2}}{a}, \quad \text{where } -a<x<a$	
(c) **[2]**	$\text{f}(-x) = \dfrac{-x\sqrt{(-x)^2+a^2}}{a}$ $= -\dfrac{x\sqrt{x^2+a^2}}{a}$ $= -\text{f}(x) \text{ (shown)}$  	
(d) **[3]**	$\text{Area} = 2\int_0^A \dfrac{x\sqrt{x^2+a^2}}{a}\,\text{d}x \quad \text{(by symmetry)}$ $= \dfrac{1}{a}\int_0^A 2x\sqrt{x^2+a^2}\,\text{d}x$ $= \left(\dfrac{1}{a}\right)\left[\dfrac{2}{3}\left(x^2+a^2\right)^{\frac{3}{2}}\right]_0^A$ $= \dfrac{2}{3a}\left(\left(A^2+a^2\right)^{\frac{3}{2}} - a^3\right)$	

3.6 A complex number z varies with t such that

$$z = 2\cos t + \mathrm{i}(3\sin t), \quad \text{where } 0 \leqslant t < 2\pi.$$

(a) By taking $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$, sketch on an Argand diagram the curve that shows the positions of the points representing the complex number z. Find the cartesian equation of this curve. [3]

[DHS 2025 P1 Q8]

Solution		**Comments**
(a) **[3]**	 $x = 2\cos t, \quad y = 3\sin t$ $\cos t = \frac{x}{2}, \quad \sin t = \frac{y}{3}$ $\cos^2 t + \sin^2 t = 1$ $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$	

3.7 A curve C has equation $y=x+\dfrac{9}{x+1}$.

(a) Using differentiation, find the range of values of x such that C concaves upwards. [2]

(b) Sketch C. [3]

[DHS 2025 P2 Q1]

Solution		Comments
(a) **[2]**	$y=x+\dfrac{9}{x+1}$ $\dfrac{dy}{dx}=1-\dfrac{9}{(x+1)^2}$ $\dfrac{d^2y}{dx^2}=\dfrac{18}{(x+1)^3}$ C concaves upwards $\Rightarrow \dfrac{18}{(x+1)^3}>0$ $(x+1)^3>0 \Rightarrow x>-1$	
(b) **[3]**	$\dfrac{dy}{dx}=0$: $(x+1)^2=9$ $x+1=\pm 3 \Rightarrow x=2$ or $x=-4$ Stationary points: $(-4,-7)$, $(2,5)$ $x=0$: $y=9$ Asymptotes: $x=-1$, $y=x$ 